CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M.Tech. (CSE) Sem - I Examination, 2010-11
EC – 702 Digital Communications & Coding

Date: 04.01.11, Tuesday **Time: 10.30 a.m to1.30 p.m** **Maximum Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Answer to the two sections must be written in the separate answer books.**
3. **Figures to the right indicate *full* marks.**

SECTION- I

Q.1

A	How do you represent band pass signal?	3
B	Which are the properties of quadrature components?	3
C	How do you design signal for band limited channel?	3
D	What is Matched filter demodulator?	3

Q.2

A	Explain probability of error for binary modulation.	6
B	Explain Memory less Modulation Method for PAM Signals	5

OR

A	How do you design band limited signal with controlled ISI?	6
B	What is probability ?Explain Conditional Probabilities	5

Q.3

A	Explain with block diagram binary PSK receiver	6
B	What is binomial distribution?	6

OR

A	Explain Costas loop with block diagram	6
B	Explain probability of error for envelope detection of correlated binary signals.	6

SECTION – II

Q4

A A signal amplitude x is a random variable uniformly distributed in the range (-1,1). This signal is passed through an amplifier of gain 2.The output y is also a common variable, uniformly distributed in the range (-2,2). Determine entropies H(x) and H(y) **5**

B Design the code tree and state diagram for coder given as N=3 & υ =2 where $\upsilon_1 = s_1 \oplus s_2 \oplus s_3$, $\upsilon_2 = s_1 \oplus s_3$ **6**

Input data sequence is 11010

Q.5

A Find the generator polynomial for g(x) for a (7,4) cyclic code and find code vector for the following data vector 1010, 1111, 0001and 1000. **6**

B A zero memory source emits six messages with probabilities 0.3, 0.25, 0.15, 0.12, 0.1 and 0.08. Find the 4-ary Huffman code. Determine its average word length, the efficiency and the redundancy **6**

OR

A Explain decision feedback equalizer with block diagram **6**

B Explain Linear adaptive equalizer based on the MSE criterion **6**

Q.6

A Explain a discrete-time model for a channel with ISI **6**

B Explain multi channel digital communication in AWGN channel **6**

OR

A What is Tomlinson harashima Precoding? **6**

B Explain multi carrier communication system with block diagram. **6**